

DTFT $\rightarrow$ can be applied to discrete time signal

Time domain $\xrightarrow{\text{DTFT}}$ frequency domain

$$x[n] \xrightarrow{\text{DTFT}} X(\omega) \text{ or } X(e^{j\omega})$$

DTFT:

$$X(\omega) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$$

Z-transform:

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

here $z = e^{j\omega}$

Inverse property (IDTFT):

$$X(\omega) \xrightarrow{\text{IDTFT}} x(n)$$

$$x(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\omega) e^{j\omega n} d\omega$$

DTFT applications:

(1) We are trying to find the magnitude and phase

We could have $X(\omega)$ in (1) polar (2) rectangular form

$$X(\omega) = |X(\omega)| \angle X(\omega)$$

↓ ↓
Magnitude phase

$$X(\omega) = X_R(\omega) + jX_I(\omega)$$

$$|X(\omega)| = \sqrt{X_R^2(\omega) + X_I^2(\omega)}$$

$$\angle X(\omega) = \tan^{-1} \left[\frac{X_I(\omega)}{X_R(\omega)} \right]$$

- Used in speech, signal, image processing etc.

Simple example:

$$(1) \quad x(n) = 1; \quad 0 \leq n \leq 4$$

$$0; \quad \text{otherwise}$$

Find DTFT.

So,

$$X(\omega) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$$

$$= 1 \times 1 + 1 \times e^{-j\omega} + 1 \times e^{-2j\omega} + 1 \times e^{-3j\omega} + 1 \times e^{-4j\omega}$$

$$= 1 + e^{-j\omega} + e^{-2j\omega} + e^{-3j\omega} + e^{-4j\omega}$$

$$= \frac{1(e^{-j\omega \times 5} - 1)}{e^{-j\omega} - 1}$$

$$= \frac{e^{-j\omega 5} - 1}{e^{-j\omega} - 1}$$

$$= \frac{e^{-j\frac{\omega 5}{2}} (e^{-j\frac{\omega 5}{2}} - e^{j\frac{\omega 5}{2}})}{e^{-j\frac{\omega}{2}} (e^{-j\frac{\omega}{2}} - e^{j\frac{\omega}{2}})}$$

$$\left[\frac{e^{j\theta} - e^{-j\theta}}{2j} = \sin\theta \right]$$

$$= e^{-2j\omega} \times \frac{\sin \frac{5\omega}{2}}{\sin \frac{\omega}{2}} \quad [re^{j\theta}]$$

$$\angle X(\omega)$$

$$= -2\omega$$

$$|X(\omega)|$$

Q2 FT of signal $x(n) = a^{|n|}$; $-1 < a < 1$

$\Downarrow$

$$x(n) = a^n; \quad n \geq 0$$

$$x(n) = a^{-n}; \quad n < 0$$

$$\therefore X_1(\omega) = \sum_{n=-\infty}^{\infty} x_1(n) e^{-j\omega n}$$

$$= \sum_{n=0}^{\infty} a^n e^{-j\omega n}$$

$$= \sum_{n=0}^{\infty} [a e^{-j\omega}]^n$$

$$= \frac{1}{1 - a e^{-j\omega}}$$

[As a is a fraction, Σ converges]

$$X_2(\omega) = \sum_{n=-\infty}^{-1} a^{-n} e^{-j\omega n}$$

$$= \sum_{n=1}^{\infty} a^n e^{j\omega n} \quad [\text{sign changed}]$$

$$= \frac{a e^{j\omega}}{1 - a e^{j\omega}}$$

$$X(\omega) = X_1(\omega) + X_2(\omega)$$

$$= \frac{1}{1 - a e^{-j\omega}} + \frac{a e^{j\omega}}{1 - a e^{j\omega}}$$

$$= \frac{1 - a e^{j\omega} + a e^{j\omega} (1 - a e^{-j\omega})}{(1 - a e^{-j\omega})(1 - a e^{j\omega})}$$

$$= \frac{1 - \cancel{a e^{j\omega}} + \cancel{a e^{j\omega}} - a^2}{1 - a e^{j\omega} - a e^{-j\omega} + a^2}$$

$$= \frac{1 - a^2}{1 - a(e^{j\omega} + e^{-j\omega}) + a^2}$$

$$\left[\frac{e^{j\theta} + e^{-j\theta}}{2} = \cos\theta \right]$$

$$= \frac{1 - a^2}{1 - 2a \cos\omega + a^2}$$

$$\nearrow |X(\omega)|$$

$$\angle X(\omega) = 0^\circ$$

Q3. Find $X(\omega)$ if $x(n) = \frac{1}{2} \left[\left(\frac{1}{2}\right)^n + \left(\frac{1}{4}\right)^n \right] u[n]$

$$\text{So, } x(n) = \frac{1}{2} \left[\left(\frac{1}{2}\right)^n + \left(\frac{1}{4}\right)^n \right] ; n \geq 0$$

$$= 0 ; n < 0$$

$$X(\omega) = \sum_{n=0}^{\infty} \frac{1}{2} \left[\left(\frac{1}{2}\right)^n + \left(\frac{1}{4}\right)^n \right]$$

$$1 - ae^{-j\omega} = 1 - a\cos\omega + ja\sin\omega$$

$$|1 - ae^{-j\omega}| = \sqrt{(1 - a\cos\omega)^2 + (a\sin\omega)^2}$$

$$= \sqrt{1 - 2a\cos\omega + a^2\cos^2\omega + a^2\sin^2\omega}$$

$$= \sqrt{1 - 2a\cos\omega + a^2}$$

DFT

⇒ Discrete Fourier Transform



$$\text{Hence, } X[k] = \sum_{n=0}^{N-1} x[n] \cdot e^{-j \times \frac{2\pi k}{N} n}$$

$$k: 0 \text{ to } N-1$$

$$n: 0 \text{ to } N-1$$

Ex:

$$(1) \quad x[n] = a^n u[n]$$

n from 0 to $N-1$

$$\begin{aligned} \therefore X[k] &= \sum_{n=0}^{N-1} x[n] \cdot e^{-j \times \frac{2\pi k}{N} n} \\ &= \sum_{n=0}^{N-1} a^n \cdot e^{-j \times \frac{2\pi k}{N} n} \end{aligned}$$

ARK slides

Lecture -1 :

For a continuous time periodic signal,

$$x(t + T_0) = x(t) \quad \text{where} \quad \omega_0 = \frac{2\pi}{T_0}$$

So, equation of k th harmonic or how much does the frequency for that sample contribute,

$$c_k = \frac{1}{T_0} \int_{T_0} x(t) e^{-jk\omega_0 t} dt = \int_{t_0}^{t_0+T_0} x(t) e^{-jk\omega_0 t} dt$$

$$\text{where, } x(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t}$$

Multiplying $x(t)$ by $e^{-jk\omega_0 t} \rightarrow$ the curve now acts periodically,

each value becomes a radius and $k\omega_0$ an angle, k is

discrete so $k\omega_0$ repeats the angle

c_k = center of mass of the wound curve

$$\text{Ex: } x(t) = 1 + 0.8 \cos(\omega_0 t) + 0.5 \cos(2\omega_0 t)$$

It has a DC component = 1

a cosine at ω_0 , a cosine at $2\omega_0$

So, k 's values appear from $k=0, \pm 1, \pm 2$ and none else

We know,

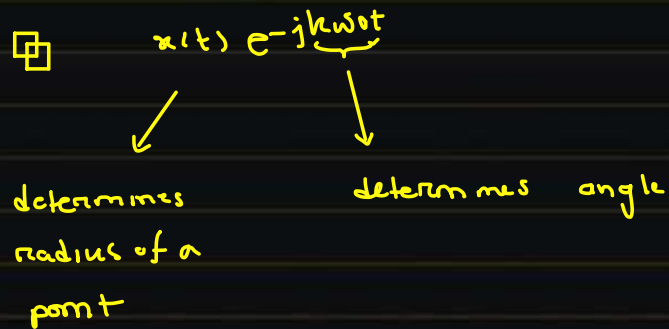
$$c_k = \frac{1}{T_0} \int_{T_0} x(t) e^{-jk\omega_0 t} dt$$

here, $e^{-jk\omega t}$ is the rotating complex exponential

So, As t increases, it rotates around the complex plane thanks to $k\omega$

When k goes out of range, hence $k=3$,

its center of mass gets cancel, hence $c_3 = 0$



* So as time moves forward, we are winding / drawing signal around the origin

So, $c_k = \frac{1}{T_0} \int_{T_0} x(t) e^{-jk\omega t} dt$ is the centroid of the curve

□ $k=1$ gives $c_1 = 0.4$

$$x(t) = 1 + 0.8 \cos(\omega t) + 0.5 \cos(2\omega t)$$

$$= 1 + 0.8 \times \frac{e^{j\omega t} + e^{-j\omega t}}{2} + 0.5 \times \frac{e^{j2\omega t} + e^{-j2\omega t}}{2}$$

$$= 1 + 0.4 e^{j\omega t} + 0.4 e^{-j\omega t} + 0.25 e^{j2\omega t} + 0.25 e^{-j2\omega t}$$



So, $k=1$ gives $c_1 = 0.4$

$k=2$ gives $c_2 = 0.25$

- As our signal doesn't have a 3rd harmonic, different parts of signal curve cancel out one another

$$\therefore c_3 = 0$$

Because the signal is periodic $\rightarrow$

$$x(t + T_0) = x(t)$$

$$\text{Rotating exponential} = e^{-jk\omega_0 t}$$

$$\begin{aligned} \text{For } t + T_0 &\rightarrow e^{-jk\omega_0(t+T_0)} \\ &= e^{-jk\omega_0 t} \times e^{-jk\omega_0 T_0} \end{aligned}$$

$$\text{We know, } \omega_0 = \frac{2\pi}{T_0}$$

$$\therefore \omega_0 T_0 = 2\pi$$

$$= e^{-jk\omega_0 t} \times e^{-jk2\pi} = e^{-jk\omega_0 t}$$

$\downarrow$

$$\cos(2\pi k) - j \sin(2\pi k)$$

$$1 - 0 \times j = 1$$

So, for $t + T_0$, we get the same rotation

WE ONLY NEED TO CHECK FOR ONE PERIOD

□ UNIFORM SAMPLING OF A CTPS gives us DTS

Suppose we have $x(t + T_0) = x(t)$

We now want to make it using discrete samples

(1) Choose a sampling period $\rightarrow T_s$

So, now instead of looking continuously, we look at

the signal at every $t = nT_s$ where $n \in \mathbb{Z}$

So, we can say,

$$x[n] = x(nT_s)$$



$$x(t) \rightarrow x[n]$$

□ Let each time period of continuous signal contains N samples

$$\therefore T_0 = NT_s$$

that is, if $T_s = 1\text{ s}$ and sampling period $= 0.25\text{ s}$

$$\therefore N = \frac{T_0}{T_s} = 4$$

So, samples in one period of DTS: $x[0], x[1], x[2], x[3]$

Show that if $x(t+T_0) \rightarrow x(t)$, then $x[n+N] \rightarrow x[n]$

We know,

$$x[n] = x(nT_s)$$

$$\begin{aligned}\therefore x[n+N] &= x((n+N)T_s) \\ &= x(nT_s + NT_s)\end{aligned}$$

$$\text{but } NT_s = T_0$$

$$\therefore x[n+N] = x(nT_s + T_0)$$

As $x(t)$ is periodic

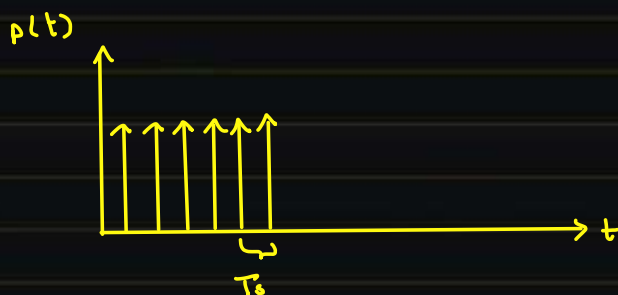
$$x(nT_s + T_0) = x(nT_s)$$

$$\therefore x[n+N] = x(nT_s) = x[n] //$$

□ The SAMPLING IMPULSE TRAIN

$$p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$$

↓
Simply means put an impulse every T_s sections



Multiplying $x(t)$ by $p(t)$, we are extracting the values of $x(t)$ after each T_s spacing across all domain.

So, $x_p(t) = x(t)p(t)$ → IMPULSE SIGNAL
↓
only contains information of $x(t)$ at $t = nT_s$

So, it is made up of impulses



$$\therefore x_p(t) = x(t)p(t)$$

$$= x(t) \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$$

Sifting / Sampling property of the delta function

$$\int_{-\infty}^{\infty} f(t) \delta(t - a) dt = f(a)$$

A very useful related property is:

$$f(t)\delta(t-a) = f(a)\delta(t-a)$$

This means when a function multiplies a delta impulse, the function can be replaced by its value at the impulse's location.

For example:

$$t^2\delta(t-3)$$

Since the impulse only exists at $t = 3$:

$$t^2\delta(t-3) = 3^2\delta(t-3)$$

Therefore:

$$t^2\delta(t-3) = 9\delta(t-3)$$

So by this property,

$$x(t)\delta(t-nT_s) = x(nT_s)\delta(t-nT_s) \\ \downarrow \\ x[n]$$

$$\therefore x_p(t) = \sum_{n=-\infty}^{\infty} x[n]\delta(t-nT_s) \\ \downarrow$$

So, each sample becomes weight of an impulse

Ex, if $x[0] = 2$, $x[1] = 5$, $x[2] = -1$

$$\therefore x_p(t) = 2\delta(t) + 5\delta(t-T_s) - \delta(t-2T_s)$$



AS SAMPLES $\rightarrow$ WEIGHTED IMPULSES NOW

We know,

For the sampled version of $x(t) = x_p(t)$,

$$C_n = \frac{1}{T_0} \int_{t_0}^{T_0+t_0} x_p(t) e^{-jk\omega_0 t} dt \\ = \frac{1}{T_0} \int_{T_0}^{\infty} \left\{ \sum_n x[n]\delta(t-nT_s) \right\} e^{-jk\omega_0 t} dt$$

n not dependent on t

$$c_k = \frac{1}{T_0} \sum_n x[n] \underbrace{\int_{T_0} \delta(t-nT_s) e^{-jk\omega_0 t} dt}_{e^{-jk\omega_0 nT_s}}$$

$$= \frac{1}{T_0} \sum_n x[n] e^{-jk\omega_0 nT_s}$$

Over one period, we only need N samples

BEFORE MOVING,

$$\text{CTFS coefficient, } c_k = \frac{1}{T_0} \int_{t_0}^{t_0+T_0} x_p(t) e^{-jk\omega_0 t} dt$$

So, interval T_0 , but it doesn't have to start at 0.

if we begin from 0, we get the last one $N T_s$, which is T_0 ,

the same point for periodic signal

So, we choose $t_0 = -\frac{T_s}{2}$ instead of 0.

Now, 0 to $(N-1)T_s$ is inside t_0 to t_0+T_0

Back to:

$$c_k = \frac{1}{T_0} \sum_n x[n] e^{-jk\omega_0 nT_s}$$

$$= \frac{1}{T_0} \sum_{n=0}^{N-1} x[n] e^{-jk\omega_0 nT_s}$$

We know,

$$\begin{aligned} \omega_0 &= \frac{2\pi}{T_0} \\ &= \frac{2\pi}{NT_s} \end{aligned}$$

$$\therefore \omega_s T_s = \frac{2\pi}{N}$$

$$\therefore C_k = \frac{1}{T_s} \sum_{n=0}^{N-1} x[n] e^{-jk \frac{2\pi}{N} n}$$



FINITE SUM instead OF INTEGRAL AFTER SAMPLING

□ ONLY N DISTINCT COEFFICIENTS

We now have,

$$C_k = \frac{1}{T_s} \sum_{n=0}^{N-1} x[n] e^{-jk \times \frac{2\pi}{N} \times n}$$

For,

$$C_{k+N} = \frac{1}{T_s} \sum_{n=0}^{N-1} x[n] e^{-j \times \frac{2\pi}{N} \times (k+N)n}$$

$$\begin{aligned} \text{Now, } e^{-j \times \frac{2\pi}{N} \times (k+N)n} &= e^{-j \times \frac{2\pi}{N} \times k \times n} \times e^{-j \times \frac{2\pi}{N} \times N \times n} \\ &= e^{-jk \times \frac{2\pi}{N} n} \times e^{-j 2\pi n} \\ &\quad \downarrow \\ &\quad 1 \text{ always} \\ &= e^{-jk \times \frac{2\pi}{N} n} \end{aligned}$$

$$\therefore \boxed{C_{k+N} = C_k}$$



So the co-efficients ARE ALSO PERIODIC with N.

$$\text{So, } C_N = C_0$$

$$C_{N+1} = C_1$$

$$C_{N+2} = C_2$$

...

So, only $C_0, C_1, \dots, C_{N-1}$ are unique

Orthogonality trick:

We know,

$$c_k = \frac{1}{T_0} \sum_{m=0}^{N-1} x[m] e^{-j \times \frac{2\pi}{N} \times km}$$

$-j$ used in ANALYSIS, $+j$ used in SYNTHESIS

Now, $T_0 c_k = \sum_{m=0}^{N-1} x[m] e^{-j \times \frac{2\pi}{N} \times km}$

$$\text{or, } \sum_{k=0}^{N-1} T_0 c_k e^{j \frac{2\pi}{N} kn} = \sum_{k=0}^{N-1} \left(x[m] e^{-j \times \frac{2\pi}{N} \times km} \right) e^{j \frac{2\pi}{N} \times kn}$$

$$\text{or, } \sum_{k=0}^{N-1} (T_0 c_k) e^{j \times \frac{2\pi}{N} \times kn} = \sum_{m=0}^{N-1} x[m] \sum_{k=0}^{N-1} e^{j \times \frac{2\pi}{N} \times k(n-m)}$$

$$\text{or, } \sum_{k=0}^{N-1} (T_0 c_k) e^{j \frac{2\pi}{N} kn} = x[n] \sum_{k=0}^{N-1} \left\{ e^{j \frac{2\pi}{N} (n-m)} \right\}^k$$

if $n-m=0$,

N

if $n \neq m$,

$$\frac{1 - e^{j \frac{2\pi}{N} (n-m)N}}{1 - e^{j \frac{2\pi}{N} (n-m)}} = \frac{1 - 1}{1 - e^{j \frac{2\pi}{N} (n-m)}} = 0$$

So, synthesis only exists for $n=m$

So,

$$\sum_{k=0}^{N-1} (T_0 c_k) e^{j \frac{2\pi}{N} kn} = N x[n]$$

$$\therefore x[n] = \frac{1}{N} \sum_{k=0}^{N-1} (T_0 c_k) e^{j \frac{2\pi}{N} kn}$$

Discrete Fourier Series (DFS)

Suppose we have an N -periodic sequence $\rightarrow \tilde{x}[n]$

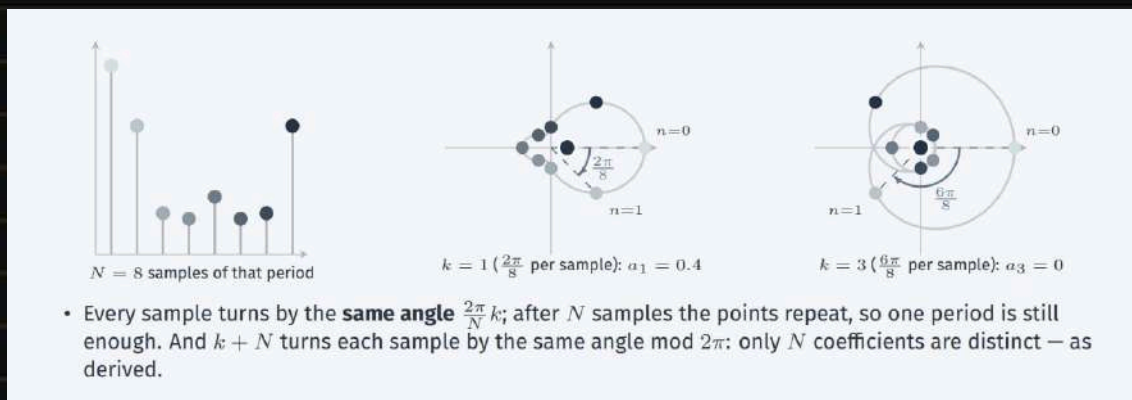
Analysis

Discrete Fourier series co-efficients,

$$a_k = \frac{1}{N} \sum_{n=0}^{N-1} \tilde{x}[n] e^{-j \times \frac{2\pi}{N} \times kn}$$

Synthesis :

$$\tilde{x}[n] = \sum_{k=0}^{N-1} a_k e^{j \frac{2\pi}{N} \times kn}$$



Dissection:

$$a_k = \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-j \times \frac{2\pi}{N} \times kn}$$

$$\text{so, each term} = x[n] e^{-j \times \frac{2\pi}{N} \times kn}$$

the n th sample radius = $x[n]$

rotates by an angle = $-\frac{2\pi}{N} kn$

so, each sample becomes a point on complex plane

5. Why divide by N ?

After rotating all the samples, we add them:

$$\sum_{n=0}^{N-1} x[n] e^{-j(2\pi/N)kn}$$

Then divide by N :

$$a_k = \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-j(2\pi/N)kn}$$

Why?

Because we're finding the average location / center of mass of the N rotated points.



So:

$$a_k = \text{center of mass of the wound samples}$$

This is exactly analogous to the CTFS coefficient being the center of mass of the continuously wound curve.

Lecture 1 - DTFT and DFT

k decides wind rate
↓
frequency

* angle of rotation betw two consecutive samples = $-\frac{2\pi k}{N}$

* winding factor = $e^{-j \times \frac{2\pi}{N} kn}$

if $N=8, k=1,$

$$\text{winding factor} = e^{-j \frac{2\pi}{8} n} = e^{-j \frac{\pi}{4} n}$$

$$\text{for } n=0, \text{ winding factor} = 1 = e^{-j0}$$

$$\text{for } n=1, \text{ " " " } = e^{-j \frac{\pi}{4}}$$

$$\text{So, } \Delta\theta = -\frac{\pi}{4} - 0 = -\frac{\pi}{4}$$

Not the original angle betw complex representation of two consecutive samples btw

$$\text{Ex: } x[0] = 2 = 2 + j0$$

$$x[1] = 3 = 3 + j0$$

The angle is $-\frac{\pi}{4}$ because its angle betn

$$x[0] e^{-j0} \quad \text{and} \quad x[1] e^{-j\frac{\pi}{4}}$$

So, original sample $\xrightarrow{x e^{-j\frac{2\pi}{N}kn}}$ wound sample

DFS ANALYSIS:

$$\text{DFS co-efficient} \rightarrow a_k = \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-j \times \frac{2\pi}{N} \times kn}$$

↓ DFT

Discrete Fourier transform $\rightarrow$

DFT

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j \frac{2\pi}{N} \times kn}$$

$$\text{So, } X[k] = N a_k$$

TWIDDLE FACTOR: represents rotation in complex plane

$$W_N = e^{-j \frac{2\pi}{N}}$$

$$W_N^k = e^{-j \frac{2\pi}{N} k}$$

W_N is the fundamental twiddle factor for N-point DFT

k determines how many fundamental rotations we make $\Rightarrow W_N^k$

Think of W_N as one step around a circular track:

$$W_N = \text{one basic step}$$

Then:

$$W_N^k = k \text{ steps}$$

And:

$$W_N^{nk} = n \times k \text{ steps}$$

So the most important identity to memorize is:

$$W_N^k = e^{-j2\pi k/N}$$

and consequently:

$$W_N^{nk} = e^{-j2\pi nk/N}$$

So, now, we know,

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j \frac{2\pi}{N} \times kn}$$

fundamental / base twiddle factor = W_N

$$k\text{th} = W_N^k$$

$$e^{-j \times \frac{2\pi}{N} \times kn} = (e^{-j \frac{2\pi}{N} k})^n = W_N^{nk}$$

$$\text{So, } X[k] = \sum_{n=0}^{N-1} x[n] W_N^{nk} \quad \text{where } k=0 \dots N-1$$

IDFT

Inverse DFT: SYNTHESIS

$$\begin{aligned} x[n] &= \frac{1}{N} \sum_{k=0}^{N-1} X[k] W_N^{-kn} \\ &= \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{-j \frac{2\pi}{N} kn} \end{aligned}$$

You should be able to write these immediately:

DFT

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}$$

IDFT

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j2\pi kn/N}$$

Difference

Forward:

$$-j$$

Inverse:

$$+j$$

And the $1/N$ is conventionally placed in the inverse.

THE ENTIRE DERIVATION IN ONE CHAIN

* We began with $x(t) \rightarrow$ a CTPS

We say $x(t + T_0) = x(t)$ where $T_0 =$ one period

For CTPS,

$$\omega_0 = \frac{2\pi}{T_0}$$

analysis, $c_k = \frac{1}{T_0} \int_{t_0}^{t_0+T_0} x(t) e^{-jk\omega_0 t} dt$

synthesis, $x(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t}$

Now for continuous to discrete:

we only retain information after every T_s seconds

let there are total N samples we retaining information

for, so, $T_0 = NT_s$

$$\text{also, } x[n] = x(nT_s)$$

↳ value at nT_s particularly

$$\text{As, } x(t + T_0) = x(t)$$

$$\text{then } x[n+N] = x[n]$$

Using impulse train,

$$\text{impulse signal, } x_p(t) = x(t)p(t)$$

$$= x(t) \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$$

$$= \sum_{n=-\infty}^{\infty} x(t) \delta(t - nT_s)$$

$$[f(t) \delta(t-a) = f(a) \delta(t-a)]$$

$$[\int_{-\infty}^{\infty} f(t) \delta(t-a) dt = f(a)]$$

$$= \sum_n x(nT_s) \delta(t - nT_s)$$

$$= \sum_n \downarrow x[n] \delta(t - nT_s)$$

$$\text{So, we have, } x_p(t) = \sum_n x[n] \delta(t - nT_s)$$

$$\text{Now, } C_n = \frac{1}{T_0} \int_{T_0} x_p(t) e^{-jk\omega_0 t} dt$$

$$= \frac{1}{T_0} \int_{T_0} \left(\sum_n x[n] \delta(t - nT_s) \right) e^{-jk\omega_0 t} dt$$

$$= \frac{1}{T_0} \sum_n x[n] \int \delta(t - nT_s) e^{-jk\omega_0 t} dt$$

$$= \frac{1}{T_0} \sum_n x[n] \int e^{-jk\omega_0 nT_s}$$

$$= \frac{1}{T_0} \sum_n x[n] e^{-jk\omega_0 nT_s}$$

$$\omega_0 T_s = \frac{2\pi}{N}$$

$$\therefore c_k = \frac{1}{T_0} \sum_n x[n] e^{-j k \times \frac{2\pi}{N} x n}$$

As total samples in one period N and $x[n+N] = x[n]$

No. of unique samples = N

$$\text{So, } c_k = \frac{1}{T_0} \sum_{n=0}^{N-1} x[n] e^{-j k \times \frac{2\pi}{N} x n}$$

$$* c_{k+N} = c_k$$

$$\text{We get, } x[n] = \sum_{k=0}^{N-1} (T_0 c_k) e^{j k \times \frac{2\pi}{N} x n} \rightarrow \text{synthesis}$$

For discrete Fourier series,

$$a_k = \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-j k \times \frac{2\pi}{N} x n} \rightarrow \text{analysis}$$

$$\text{DFT, } X[k] = \sum_{n=0}^{N-1} x[n] e^{-j k \times \frac{2\pi}{N} x n}$$

$$\text{IDFT, } x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{-j k \times \frac{2\pi}{N} x n}$$

Example Question

Given the 4-point sequence

$$x[n] = \{1, 2, 3, 4\}, \quad n = 0, 1, 2, 3$$

(a) Find its 4-point DFT $X[k]$.

(b) Find $x[n]$ back using the IDFT.

(c) Explain the meaning of W_4 .

(a) For 4-point DFT, we need W_4

$$W_4 = e^{-j \times \frac{2\pi}{4}} = e^{-j \frac{\pi}{2}} = \cos \frac{\pi}{2} - j \sin \left(\frac{\pi}{2} \right) = -j$$

$$W_4^0 = (-j)^0 = 1$$

$$W_4^1 = (-j)^1 = -j$$

$$W_4^2 = (-j)^2 = -1$$

$$W_4^3 = (-j)^3 = -j \times (-j) \times (-j) = j$$

Now,

$$X[k] = \sum_{n=0}^{N-1} x[n] W_4^{nk}$$

$$X[0] = \sum_{n=0}^{N-1} x[n] \times 1$$

$$= x[0] + x[1] + x[2] + x[3]$$

$$= 1 + 2 + 3 + 4 = 10 \rightarrow \text{DC comp}$$

$$X[1] = \sum_{n=0}^{N-1} x[n] W_4^n$$

$$= x[0] W_4^0 + x[1] W_4^1 + x[2] W_4^2 + x[3] W_4^3$$

$$= 1 \times 1 + 2(-j) + 3 \times (-1) + 4j$$

$$= -2 + 2j$$

$$X[2] = x[0] W_4^0 + x[1] W_4^2 + x[2] W_4^4 + x[3] W_4^6$$

$$= 1 \times 1 + 2 \times (-1) + 3 \times 1 + 4(-1)$$

$$= 4 - 2 - 4 = -2$$

$$X[3] = x[0] W_4^0 + x[1] W_4^3 + x[2] W_4^6 + x[3] W_4^9$$

$$= 1 + 2j - 3 - 4j$$

$$= -2 - 2j$$

$$X[k] = \{10, -2 + 2j, -2, -2 - 2j\}$$

$$(b) \quad \text{IDFT} \rightarrow x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] W_4^{-nk}$$

$$x[0] \rightarrow \frac{1}{4} \sum_{k=0}^3 x[k]$$

Now,

$$x[0] = \frac{1}{4} (x[0] + x[1] + x[2] + x[3])$$

$$= \frac{1}{4} (10 - 2 + 2j - 2 - 2 - 2j)$$

$$= 1$$

$$x[1] \rightarrow \frac{1}{4} \sum_{k=0}^3 x[k] W_4^{-k}$$

$$= \frac{1}{4} (x[0] \times 1 + x[1] \times W_4^{-1} + x[2] \times W_4^{-2} + x[3] \times W_4^{-3})$$

$$= \frac{1}{4} (10 + j(-2+2j) - 2(-1) + (-j)(-2-2j))$$

$$= \frac{1}{4} (10 - 2j - 2 + 2 + 2j - 2)$$

$$= 2$$

$$\text{Similarly, } x[2] = 3, x[3] = 4$$

$$(c) \quad W_N = e^{-j\frac{2\pi}{N}}$$

For every real valued $x[n]$,

$$x[N-k] = x^*[k]$$

$$\text{here, } x[3] = -2 - 2j$$

$$x^*[3] = -2 + 2j = x[1] = x[4-1]$$

REPRESENTATIONS OF $x[n]$

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} (T_0 c_k) e^{jk \frac{2\pi}{N} n} \rightarrow \text{CTFS}$$

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} x[k] e^{jk \frac{2\pi}{N} n} \rightarrow \text{DFS}$$

$$a_k = \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-jk \frac{2\pi}{N} n} \rightarrow x[k] = N a_k$$

$$\text{So, } x[n] = \sum_{k=0}^{N-1} a_k e^{jk \frac{2\pi}{N} n}$$

$$\text{So, } x[k] = T_0 c_k = N a_k \quad \xrightarrow{\text{for sampled periodic signal } x_p(t)}$$

$$\downarrow$$
$$\frac{1}{T_0} \sum_n x[n] e^{-jk \omega_n T_s}$$

$$\downarrow$$
$$\frac{1}{T_0} \sum_n x[n] e^{-jk \frac{2\pi}{N} n} = \frac{1}{T_0} \sum_n x[n] W_N^{nk}$$

$c_k =$ CTFS coefficient of $x_p(t)$

$\downarrow$
sampled periodic signal / impulse

$$x_{\text{sam}} = x(t) p(t)$$

$$= x(t) \sum_n \delta(t - nT_s)$$

$$= \sum_n x(t) \delta(t - nT_s)$$

$$= \sum_n x(nT_s) \delta(t - nT_s)$$

$$= \sum_n x[n] \delta(t - nT_s)$$

DTFT

Aperiodic DTS $\rightarrow$ DTFT

Now, $x[n+N] \neq x[n]$

So, we consider,

let's take an aperiodic signal $x(t)$ and look at it for T_0 sec

Within this, we will make N samples

So, $x[n] = x(nT_s)$ also, $T_0 = NT_s$

Then, $x[n] \neq 0$ for $0 \leq n \leq N-1$

DTFT

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

For N -length,

$$X(e^{j\omega}) = \sum_{n=0}^{N-1} x[n] e^{-j\omega n}$$

Sampled impulse train $\rightarrow$

$$x_p(t) = \sum_{n=-\infty}^{\infty} x[n] \delta(t - nT_s)$$

Now, for CTFT,

$$\begin{aligned} X_p(j\omega) &= \int_{-\infty}^{\infty} x_p(t) e^{-j\omega t} dt \\ &= \int_{-\infty}^{\infty} \left(\sum_{n=-\infty}^{\infty} x[n] \delta(t - nT_s) \right) e^{-j\omega t} dt \end{aligned}$$

$$= \sum_{n=-\infty}^{\infty} x[n] \int_{-\infty}^{\infty} \delta(t - nT_s) e^{-j\omega t} dt$$

$$= \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega_0 n T_s}$$

$$\text{Let } \omega_0 T_s = \omega$$

$$\therefore X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

* DTFT is 2π -periodic

$$\begin{aligned} X(e^{j(\omega+2\pi)}) &= \sum_{n=-\infty}^{\infty} x[n] e^{-j(\omega+2\pi)n} \\ &= \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} \\ &= X(e^{j\omega}) \end{aligned}$$

Inverse DTFT:

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$

DTFT gives continuous spectrum while DFT gives discrete samples of that spectrum



$$\text{DFT, } X[k] = X(e^{j\omega}) \Big|_{\omega = \frac{2\pi k}{N}}$$

DFT = discrete N samples of one 2π period of DTFT

$$\omega_k = \frac{2\pi k}{N}$$

$N \uparrow$: more closely spaced frequencies

$N \downarrow$: fewer f samples

For even N :

$$k = 0, 1, \dots, \frac{N}{2}$$

correspond to:

$$0 \rightarrow \pi$$

These are the positive-frequency side.

Then:

$$k = \frac{N}{2} + 1, \dots, N - 1$$

correspond to frequencies beyond π .

Because the DTFT wraps every 2π , these represent negative frequencies.

 physical frequency, $\Omega \text{ (rad/s)} = \frac{\omega}{T_s} \rightarrow \text{digital freq}$

We know:

$$\omega_k = \frac{2\pi k}{N}$$

Convert to physical frequency:

$$\Omega_k = \frac{\omega_k}{T_s}$$

Substitute:

$$\Omega_k = \frac{2\pi k}{NT_s}$$

The sampling angular frequency is:

$$\omega_s = \frac{2\pi}{T_s}$$

Therefore:

$$\boxed{\Omega_k = k \frac{\omega_s}{N}}$$

for the positive-frequency bins.

What about negative bins?

For:

$$k > \frac{N}{2}$$

the bin is interpreted after wrapping:

$$\omega_k = \frac{2\pi k}{N} - 2\pi$$

Factor:

$$= \frac{2\pi(k - N)}{N}$$

Thus:

$$\boxed{\Omega_k = (k - N) \frac{\omega_s}{N}}$$

So the complete rule is:

$$\Omega_k = \begin{cases} k \frac{\omega_s}{N}, & 0 \leq k \leq N/2 \\ (k - N) \frac{\omega_s}{N}, & N/2 < k \leq N - 1 \end{cases}$$

$$\Omega_k T_s = \frac{2\pi}{N} (k - N)$$

$$\text{or } \Omega_k \times \frac{2\pi}{\omega_s} = \frac{2\pi}{N} (k - N)$$

$$\therefore \Omega_k = \frac{\omega_s}{N} (k - N)$$

FFT Fast Fourier Transform (faster than DFT)

Types: (1) Inverse FFT $\Rightarrow X[k]$ given $\rightarrow$ find $x[n]$
(2) DIT FFT
(3) DIF FFT } $x[n]$ given $\Rightarrow$ find $X[k]$

Computational complexity of evaluating DFT:

$$\text{DFT} \rightarrow X[k] = \sum_{n=0}^{N-1} x[n] W_N^{nk}$$

$$W_N = e^{-j \frac{2\pi}{N}} \rightarrow \text{base twiddle factor}$$

Directly if this formula is applied

```
x = np.asarray(x, dtype=np.complex128)
N = len(x)
X = np.zeros(N, dtype=np.complex128)
for k in range(N):
    for n in range(N):
        X[k] += x[n] * np.exp(-2j * np.pi * k * n / N)
return X
```

N complex multiplications

inside them, N-1 complex additions

$\downarrow$
 $O(N^2)$

Solⁿ: FFT for computation

Two variants $\rightarrow$ (1) DIT (2) DIF $\rightarrow$ Decimation in Frequency

$\downarrow$

Decimation in Time

DIT

* Splits time domain sequence into even and odd sequences

$$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{nk} = \sum_{r=0}^{N/2-1} x[2r] W_N^{2rk} + \sum_{r=0}^{N/2-1} x[2r+1] W_N^{(2r+1)k}$$

Now,

$$W_N^{2rk} = e^{-j \times \frac{2\pi}{N} \times k \times 2r} = e^{-j \times \frac{2\pi}{N/2} \times k \times r} = W_{N/2}^{rk}$$

$$W_N^{(2r+1)k} = W_N^{2rk} \times W_N^k = W_{N/2}^{rk} \times W_N^k$$

So,

$$X[k] = \underbrace{\sum_{r=0}^{N/2-1} x[2r] W_{N/2}^{rk}}_{N/2\text{-point DFT of evens}} + W_N^k \underbrace{\sum_{r=0}^{N/2-1} x[2r+1] W_{N/2}^{rk}}_{\text{same for odds}}$$

• We give each function names.

1st part:

$$G[k] = \sum_{r=0}^{N/2-1} x[2r] W_{N/2}^{rk}$$

$$H[k] = \sum_{r=0}^{N/2-1} x[2r+1] W_{N/2}^{rk}$$

So, $X[k] = G[k] + W_N^k \cdot H[k]$

For second half, $N/2 \leq k < N$

We know,

$$G[k + N/2] = G[k]$$

$$H[k + N/2] = H[k]$$

We can use property,

$$\begin{aligned} W_N^{k+N/2} &= W_N^k \times W_N^{N/2} \\ &= W_N^k \times e^{-j \times \frac{2\pi}{N} \times \frac{N}{2}} \\ &= W_N^k \times e^{-j\pi} \\ &= W_N^k (-1) \\ &= -W_N^k \end{aligned}$$

$$\text{So, } X[k + N/2] = G[k] - W_N^k \cdot H[k]$$

Radix-2 Decimation-in-Time FFT

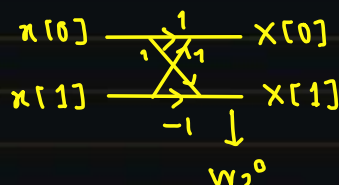
For $0 \leq k < N/2$:

$$\begin{aligned} X[k] &= G[k] + W_N^k H[k] \\ X[k + N/2] &= G[k] - W_N^k H[k] \end{aligned}$$

• Ground lvl operation is 2 point DFT

$$x[0] = g[0] + w_2^0 \cdot h[0]$$

$$x[0+1] = x[1] = g[0] - w_2^0 h[0]$$



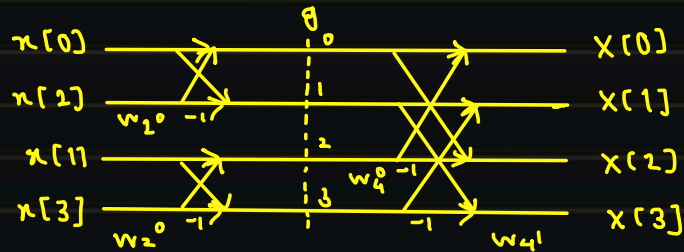
$$x[0] = x[0] + x[1]$$

$$x[1] = x[0] - x[1]$$

4-point DFT:

Bit reversal technique $\rightarrow$

<u>original</u>	<u>binary</u>	<u>reversed</u>	<u>new</u>
0	00	00	0
1	01	10	2
2	10	01	1
3	11	11	3



$$g[0] = x[0] + x[2]$$

$$g[1] = x[0] - x[2]$$

$$g[2] = x[1] + x[3]$$

$$g[3] = x[1] - x[3]$$

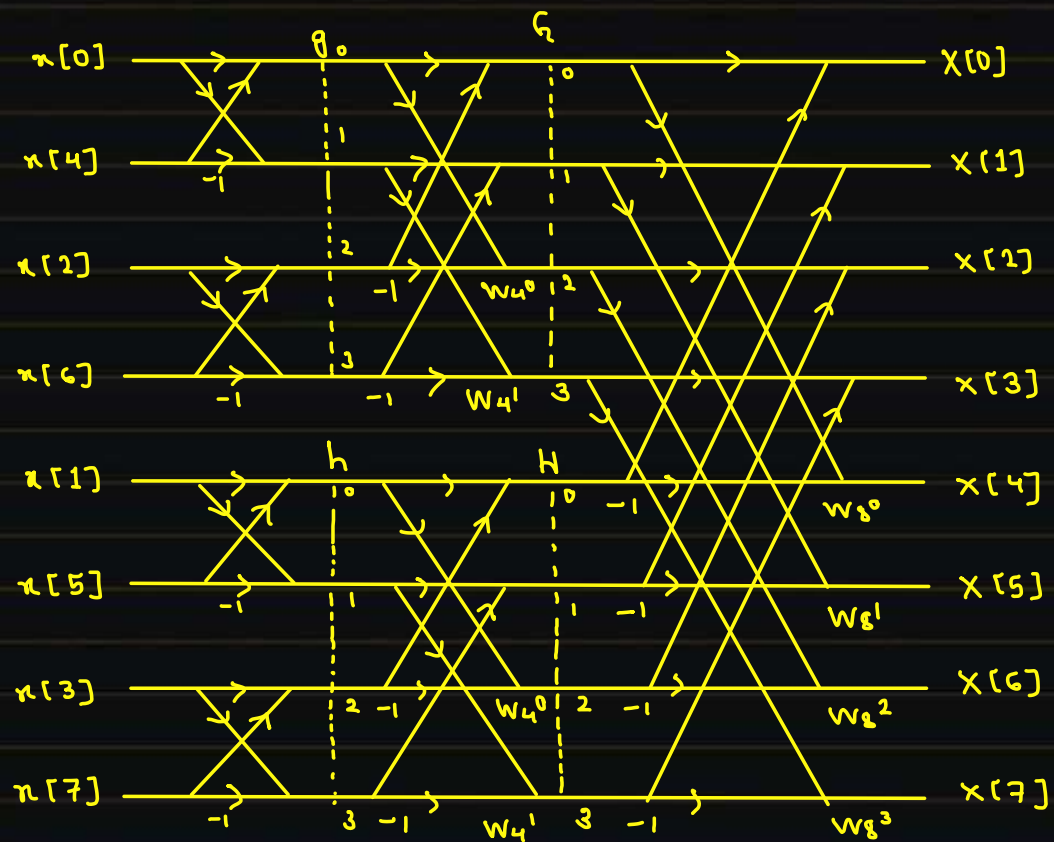
$$x[0] = g[0] + w_4^0 \times g[2]$$

$$x[1] = g[1] + w_4^1 \times g[3]$$

$$x[2] = g[0] - w_4^0 \times g[2]$$

$$x[3] = g[1] - w_4^1 \times g[3]$$

DIT 8-point FFT:



$$g[0] = x[0] + x[4]$$

$$g[1] = x[0] - x[4]$$

$$g[2] = x[2] + x[6]$$

$$g[3] = x[2] - x[6]$$

$$h[0] = x[1] + x[5]$$

$$h[1] = x[1] - x[5]$$

$$h[2] = x[3] + x[7]$$

$$h[3] = x[3] - x[7]$$

Now 2nd iteration:

$$G[0] = g[0] + W_4^0 \cdot g[2]$$

$$G[1] = g[1] + W_4^1 \cdot g[3]$$

$$G[2] = g[0] - W_4^0 \cdot g[2]$$

$$G[3] = g[1] - W_4^1 \cdot g[3]$$

$$H[0] = h[0] + W_4^0 \cdot h[2]$$

$$H[1] = h[1] + W_4^1 \cdot h[3]$$

$$H[2] = h[0] - W_4^0 \cdot h[2]$$

$$H[3] = h[1] - W_4^1 \cdot h[3]$$

Now 3rd iteration:

$$x[0] = G[0] + w_8^0 H[0]$$

$$x[4] = G[0] - w_8^0 H[0]$$

$$x[1] = G[1] + w_8^1 H[1]$$

$$x[5] = G[1] - w_8^1 H[1]$$

$$x[2] = G[2] + w_8^2 H[2]$$

$$x[6] = G[2] - w_8^2 H[2]$$

$$x[3] = G[3] + w_8^3 H[3]$$

$$x[7] = G[3] - w_8^3 H[3]$$

Pseudocode of DIT FFT:

Bit reversal array x of length $N \rightarrow O(N)$

for $s = 1$ to $\log_2 N$: $\rightarrow O(\log_2 N)$

$$M = 2^s \rightarrow O(1)$$

$$w_M = e^{-j 2\pi/M} \rightarrow O(1)$$

for $L = 0$ to $N-M$ step M : $\rightarrow O(N/M)$

$$w = 1$$

for $k = 0$ to $M/2 - 1$: $\rightarrow O(M/2)$

$$g = x[L+k] \rightarrow O(1)$$

$$h = w \cdot x[L+k+M/2] \rightarrow O(1)$$

$$x[L+k] = g+h \rightarrow O(1)$$

$$x[L+k+M/2] = g-h \rightarrow O(1)$$

$$w = w \cdot w_M \rightarrow O(1)$$

Ex:

$$N=16, M=4$$

$$L \rightarrow 0 \rightarrow 3 \rightarrow 4 \text{ times}$$

$$\frac{N}{M} = 4$$

$$\therefore, T_C = O(N/M \times M/2 \times \log_2 N)$$

$$= O(N \log_2 N)$$

Every stage has = $\frac{N}{2}$ butterflies

DIF FFT Derivation:

Instead of splitting into even and odd indices,

we simply divide into two halves

$$X[k] = \sum_{n=0}^{N/2-1} x[n] W_N^{kn} + \sum_{n=N/2}^{N-1} x[n] W_N^{k(n+N/2)}$$

$$W_N^{N/2} = -1$$

$$X[k] = \sum_{n=0}^{N/2-1} (x[n] + (-1)^k x[n+N/2]) W_N^{kn}$$

* Unlike DFT FFT,

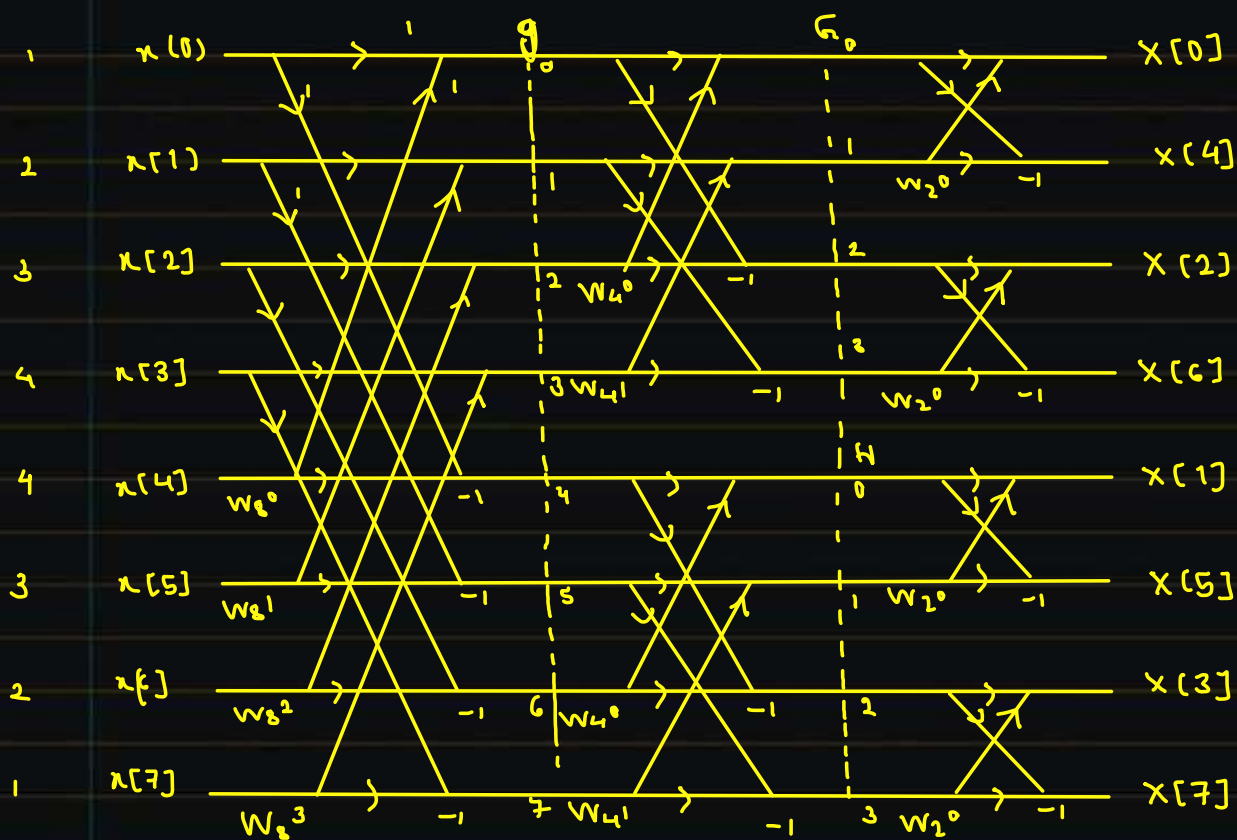
input indices appear in natural order, output in bit reversed order

Example application:

Compute DFT for sequence $\{1, 2, 3, 4, 4, 3, 2, 1\}$ using radix-2 DIF

FFT Algorithm

$$x[n] = \{1, 2, 3, 4, 4, 3, 2, 1\}$$



$$w_8^0 = 1, w_8^1 = e^{-j\pi/4} = 1/\sqrt{2} - j1/\sqrt{2}$$

$$w_8^2 = e^{-j \times \frac{2\pi}{8} \times 2} = e^{-j \times \pi/2} = -j$$

$$w_8^3 = e^{-j \times \frac{3\pi}{4}} = -1/\sqrt{2} - j1/\sqrt{2}$$

$$g[0] = x[0] + w_8^0 x[4]$$

$$g[4] = x[0] - w_8^0 x[4]$$

$$g[1] = x[1] + w_8^1 x[5]$$

$$g[5] = x[1] - w_8^1 x[5]$$

$$g[2] = x[2] + w_8^2 x[6]$$

$$g[6] = x[2] - w_8^2 x[6]$$

$$g[3] = x[3] + w_8^3 x[7]$$

$$g[7] = x[3] - w_8^3 x[7]$$

$$\text{So, } g[0] = 1 + 4 = 5$$

$$g[4] = 1 - 4 = -3$$

$$g[1] = 2 + \left(\frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}}\right) 3$$

$$g[5] = \left(2 - \frac{3}{\sqrt{2}}\right) + \frac{3}{\sqrt{2}} j$$

$$= \left(2 + \frac{3}{\sqrt{2}}\right) - \frac{3}{\sqrt{2}} j$$

$$g[2] = 3 + (-j) \times 2 = 3 - 2j$$

$$g[6] = 3 + 2j$$

$$g[3] = 4 + \left(-\frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}}\right) 1$$

$$= \left(4 - \frac{1}{\sqrt{2}}\right) - \frac{j}{\sqrt{2}}$$

$$g[7] = \left(4 + \frac{1}{\sqrt{2}}\right) + \frac{j}{\sqrt{2}}$$

2nd iteration:

$$G[0] = g[0] + W_4^0 \cdot g[1] = 5 + 3 - 2j = 8 - 2j$$

$$G[1] = g[1] + W_4^1 \cdot g[3] = \left(2 + \frac{3}{\sqrt{2}}\right) - \frac{3}{\sqrt{2}}j + (-j)g[3]$$

$$= (2 + \sqrt{2}) - (4 + \sqrt{2})j$$

$$G[2] = g[0] - W_4^0 \cdot g[2] = 5 - 3 + 2j = 2 + 2j$$

$$G[3] = g[1] - W_4^1 \cdot g[3] = (2 + \sqrt{2}) + (4 - 2\sqrt{2})j$$

$$H[0] = g[4] + W_4^0 \cdot g[6] = -3 + 3 + 2j = 2j$$

$$H[1] = g[5] + W_4^1 \cdot g[7] = (2 - \sqrt{2}) + (-4 + \sqrt{2})j$$

$$H[2] = g[4] - W_4^0 \cdot g[6] = -3 - 3 - 2j = -6 - 2j$$

$$H[3] = g[5] - W_4^1 \cdot g[7] = (2 - \sqrt{2}) + (4 + 2\sqrt{2})j$$

3rd iteration:

$$X[0] = G[0] + G[1]$$

$$X[4] = G[0] - G[1]$$

$$X[1] = H[0] + H[1]$$

$$X[5] = H[0] - H[1]$$

$$X[2] = G[2] + G[3]$$

$$X[6] = G[2] - G[3]$$

$$X[3] = H[2] + H[3]$$

$$X[7] = H[2] - H[3]$$

So, $X[0] = 10 + \sqrt{2} - (6 + \sqrt{2})j$

$$X[4] = 6 - \sqrt{2} + (2 + \sqrt{2})j$$

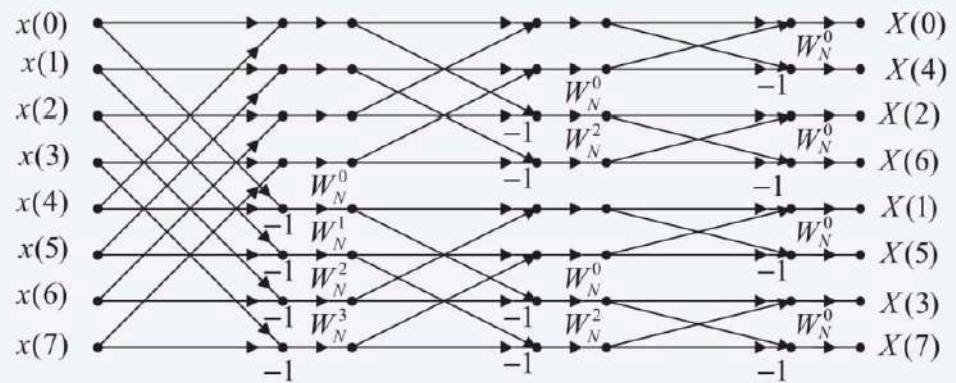
$$X[1] =$$

$$X[2] =$$

$$X[3] =$$

WRONG $\rightarrow$

- Expanding the layers gives the full 8-point DIF structure.
- Here inputs are in **natural** order, while outputs emerge in **bit-reversed** order.



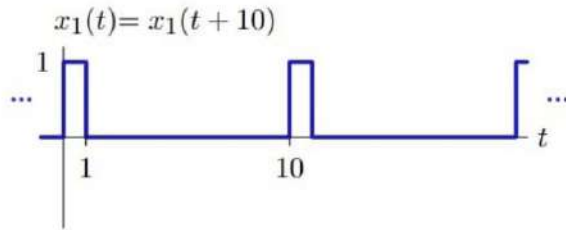
DIT IFFT

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] W_N^{-nk}$$

$$= \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{jk \times \frac{2\pi}{N} n}$$

TF question

Determine the Fourier series coefficients a_k for $x_1(t)$ shown below.



$$x_1(t) = x_1(t + T_0)$$

$$\downarrow$$

$$10$$

$$a_k = \frac{1}{T_0} \int x_1(t) e^{-jk\omega_0 t} dt$$

$$= \frac{1}{10} \int x_1(t) e^{-jk\frac{\pi}{5}t} dt$$

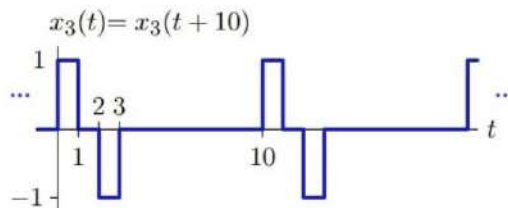
Now,

$$a_k = \frac{1}{10} \int_0^1 1 \times e^{-jk\frac{\pi}{5}t} dt + \frac{1}{10} \int_1^{10} 0 dt$$

$$= \frac{1}{10} \left[\frac{e^{-jk\frac{\pi}{5}t}}{-jk\frac{\pi}{5}} \right]_0^1$$

$$= \frac{1}{10} \left[\frac{e^{-jk\frac{\pi}{5}}}{-jk\frac{\pi}{5}} - 1 \right]$$

Determine the Fourier series coefficients c_k for $x_3(t)$ shown below.



$$a_0 = \frac{1}{10} \int_0^1 dt + \frac{1}{10} \int_1^3 (-1) dt$$

$$= 0$$

CFT:

$$x(\omega) = X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

IFT:

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} x(\omega) e^{j\omega t} d\omega$$

* FT of $\sum_{k=0}^{\infty} \alpha^k \delta(t - kT)$, $|\alpha| < 1$

$$\begin{aligned} \text{So, } x(\omega) &= \int_{-\infty}^{\infty} \left(\sum_{k=0}^{\infty} \alpha^k \delta(t - kT) \right) e^{-j\omega t} dt \\ &= \sum_{k=0}^{\infty} \alpha^k e^{-j\omega kT} \\ &= \sum_{k=0}^{\infty} (\alpha e^{-j\omega T})^k \\ &= \frac{1}{1 - \alpha e^{-j\omega T}} \end{aligned}$$

CT ques ARK

Q1)

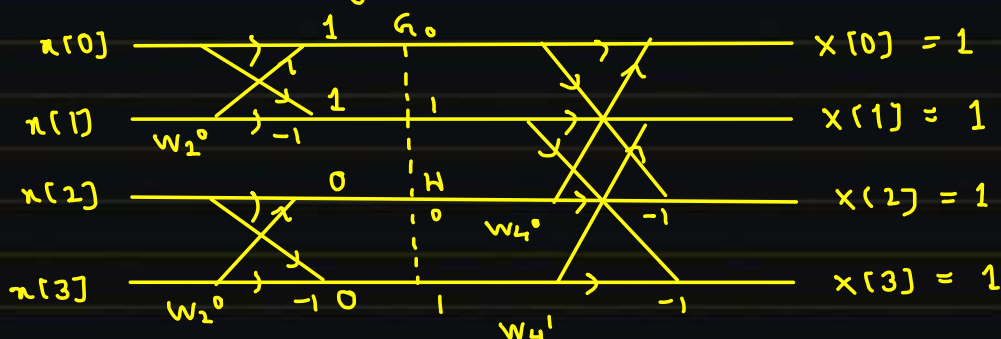
1. Consider the signal $x[n]$ defined as follows for $N = 4$:

[8]

$$x[n] = \begin{cases} 1, & \text{if } n = 0 \\ 0, & \text{if } 1 \leq n < N \end{cases}$$

(a) Compute the 4-point radix-2 DIT FFT of the signal. Draw a butterfly diagram and show your calculations on the diagram.

$$x[n] = \{1, 0, 0, 0\}$$



$$x[n] = \{1, 1, 1, 1\}$$

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-jk \times \frac{2\pi}{N} n}$$

$$= \sum_{n=0}^{N-1} x[n] W_N^{nk}$$